

Modal Logic: Tableaux Examples

2026-02-18

Examples

Example 1: $\Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ in K

1. $\models 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ Assumption

Start by assuming it's false.

1. $\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ Assumption
2. $\mathbb{T} 1, \Diamond(A \wedge B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond A \wedge \Diamond B \rightarrow \mathbb{F} 1$

False conditional means true antecedent and false consequent.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond(A \wedge B)$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond A \wedge \Diamond B$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, A \wedge B$ | $\Diamond \mathbb{T} 2$ |

True diamond introduces world 1.1.

1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$	Assumption
2.	$\mathbb{T} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, A \wedge B$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\wedge \mathbb{T} 4$
6.	$\mathbb{T} 1.1, B$	$\wedge \mathbb{T} 4$

True conjunction gives us both A and B at 1.1.

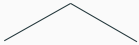
1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$	Assumption
2.	$\mathbb{T} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, A \wedge B$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\wedge \mathbb{T} 4$
6.	$\mathbb{T} 1.1, B$	$\wedge \mathbb{T} 4$
$\swarrow \quad \searrow$		
7.	$\mathbb{F} 1, \Diamond A \quad \mathbb{F} 1, \Diamond B$	$\wedge \mathbb{F} 3$
8.	$\mathbb{F} 1.1, A \quad $	$\Diamond \mathbb{F} 7$
9.	$\otimes \quad \mathbb{F} 1.2, B$	$\Diamond \mathbb{F} 7$
	$\otimes$	

When we introduce worlds that make the $\Diamond$ sentences true, both branches close. So this is valid in **K**.

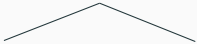
Example 2: $(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ in $\mathbf{K}$

Let's show this is a theorem in $\mathbf{K}$.

1. $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ Assumption
2. $\mathbb{T} 1, \Diamond A \vee \Diamond B \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond(A \vee B) \rightarrow \mathbb{F} 1$

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$	Assumption
2.	$\mathbb{T} 1, \Diamond A \vee \Diamond B$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
		
4.	$\mathbb{T} 1, \Diamond A \quad \mathbb{T} 1, \Diamond B$	$\vee \mathbb{T} 2$

The disjunction branches.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
		
4.	$\top 1, \Diamond A$	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\mathbb{F} 1.1, A \vee B$	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
8.	$\otimes$	$\vee \mathbb{F} 6$
		$\otimes$

Both branches close, so this **is** valid in K.

Example 3: $A \rightarrow \Box \Diamond A$ in KT

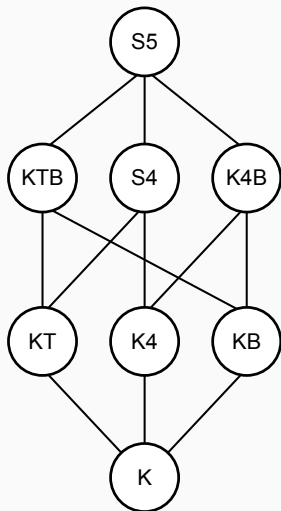
This should fail in KT. Let's show it.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, A \rightarrow \Box \Diamond A$ | Assumption |
| 2. | $\mathbb{T} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{F} 1.1, \Diamond A$ | $\Box \mathbb{F} 3$ |
| 5. | $\mathbb{F} 1.1, A$ | $\mathbb{T} \Diamond 4$ |

The tree stays open. So this is not valid in **KT**.

Note that after line 4 we've completed all the rules we needed to use in **K**, so the first four lines show it is not a theorem in **K**.

The Landscape of Modal Logics



- Stronger logics are higher in the diagram
- If a formula is **valid** in a weaker logic, it is **valid** in all stronger logics above it
- If a formula is **not valid** in a stronger logic, it is **not valid** in any weaker logic below it

Figure 1: Eight important modal logics

Example 4: $A \rightarrow \Box\Diamond A$ in KB

But this does hold in **KB**. Let's see this.

1.	$\mathbb{F} 1, A \rightarrow \Box \Diamond A$	Assumption
2.	$\mathbb{T} 1, A$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box \Diamond A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1.1, \Diamond A$	$\Box \mathbb{F} 3$
5.	$\mathbb{F} 1, A$	$\Box \Diamond 4$
	$\otimes$	

The $\Box \Diamond$ rule gives us the contradiction.

Example 5: $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ in K

This is the K axiom - it should be valid even in K.

1. $\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ Assumption
2. $\mathbb{T} 1, \Box(A \rightarrow B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box A \rightarrow \Box B \rightarrow \mathbb{F} 1$

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ | Assumption |
| 2. | $\mathbb{T} 1, \Box(A \rightarrow B)$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box A \rightarrow \Box B$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Box A$ | $\rightarrow \mathbb{F} 3$ |
| 5. | $\mathbb{F} 1, \Box B$ | $\rightarrow \mathbb{F} 3$ |

1.	$\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$	Assumption
2.	$\mathbb{T} 1, \Box(A \rightarrow B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \rightarrow \Box B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box B$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{F} 1.1, B$	$\Box \mathbb{F} 5$
7.	$\mathbb{T} 1.1, A \rightarrow B$	$\Box \mathbb{T} 2$
8.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 4$

1.	$\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$	Assumption
2.	$\mathbb{T} 1, \Box(A \rightarrow B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \rightarrow \Box B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Box B$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{F} 1.1, B$	$\Box \mathbb{F} 5$
7.	$\mathbb{T} 1.1, A \rightarrow B$	$\Box \mathbb{T} 2$
8.	$\mathbb{T} 1.1, A$	$\Box \mathbb{T} 4$
$\swarrow \quad \searrow$		
9.	$\mathbb{F} 1.1, A \quad \mathbb{T} 1.1, B$	$\rightarrow \mathbb{T} 7$
	$\otimes \quad \otimes$	

Both branches close, so this is valid.

Indeed, this is the canonical axiom for **K**.

Example 5: $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ in K

Another way to think about this is that it's saying that if you have

- $\Box(A \rightarrow B)$; and
- $\Box A$

you can derive

- $\Box B$

The general principle is that if an argument is valid, then putting a box in front of every premise and in front of the conclusion will preserve validity.

Example 6: $(\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ in K

1. $\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B)$ Assumption
2. $\mathbb{T} 1, \Box A \vee \Box B \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box(A \vee B) \rightarrow \mathbb{F} 1$

1.	$\mathbb{F} 1, (\Box A \vee \Box B) \rightarrow \Box(A \vee B)$	Assumption	
2.	$\mathbb{T} 1, \Box A \vee \Box B$	$\rightarrow \mathbb{F} 1$	
3.	$\mathbb{F} 1, \Box(A \vee B)$	$\rightarrow \mathbb{F} 1$	
4.	$\mathbb{T} 1, \Box A$	$\mathbb{T} 1, \Box B$	$\vee \mathbb{T} 2$
5.	$\mathbb{F} 1.1, A \vee B$	$\mathbb{F} 1.1, A \vee B$	$\Box \mathbb{F} 3$
6.	$\mathbb{T} 1.1, A$	$\mathbb{T} 1.1, B$	$\Box \mathbb{T} 4$
7.	$\mathbb{F} 1.1, A$	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 5$
8.	$\otimes$	$\mathbb{F} 1.1, B$	$\vee \mathbb{F} 5$
		$\otimes$	

This is valid in K. Intuitively, if either one of $\Box A$ or $\Box B$ is true, we can derive $\Box(A \vee B)$

Example 7: $(\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$ in K

This should fail; different worlds might make $\Diamond A$ and $\Diamond B$ true.

And the special case where $B = \neg A$ also suggests it is invalid.

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|----|---|----------------------------|
| 1. | $\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A \wedge \Diamond B$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond(A \wedge B)$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Diamond A$ | $\wedge \mathbb{T} 2$ |
| 5. | $\mathbb{T} 1, \Diamond B$ | $\wedge \mathbb{T} 2$ |

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{F} 1.1, A \wedge B$	$\Diamond \mathbb{F} 3$
9.	$\mathbb{F} 1.2, A \wedge B$	$\Diamond \mathbb{F} 3$

The tree stays open. World 1.1 has A but might not have B , and world 1.2 has B but might not have A .

Example 7: $\Diamond A \wedge \Diamond B \rightarrow \Diamond(A \wedge B)$ in K

This should fail - worlds accessible from 1 might be different.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A \wedge \Diamond B$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond(A \wedge B)$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Diamond A$ | $\wedge \mathbb{T} 2$ |
| 5. | $\mathbb{T} 1, \Diamond B$ | $\wedge \mathbb{T} 2$ |

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$

1.	$\mathbb{F} 1, (\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$	Assumption
2.	$\mathbb{T} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \wedge B)$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond A$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond B$	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, B$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{F} 1.1, A \wedge B$	$\Diamond \mathbb{F} 3$
9.	$\mathbb{F} 1.2, A \wedge B$	$\Diamond \mathbb{F} 3$

The tree stays open! World 1.1 has A but might not have B , and world 1.2 has B but might not have A .

Example 8: $\Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ in K

1. $\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ Assumption
2. $\mathbb{T} 1, \Box(A \vee B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box A \vee \Diamond B \rightarrow \mathbb{F} 1$

- | | | |
|----|--|-------------------------|
| 1. | $\models 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$ | Assumption |
| 2. | $\top 1, \Box(A \vee B)$ | $\rightarrow \models 1$ |
| 3. | $\models 1, \Box A \vee \Diamond B$ | $\rightarrow \models 1$ |
| 4. | $\models 1, \Box A$ | $\vee \models 3$ |
| 5. | $\models 1, \Diamond B$ | $\vee \models 3$ |

1.	$\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$	Assumption
2.	$\mathbb{T} 1, \Box(A \vee B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \vee \Diamond B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1, \Box A$	$\vee \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond B$	$\vee \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 4$
7.	$\mathbb{T} 1.1, A \vee B$	$\Box \mathbb{T} 2$
8.	$\mathbb{F} 1.1, B$	$\Diamond \mathbb{F} 5$

1.	$\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Box A \vee \Diamond B)$	Assumption
2.	$\mathbb{T} 1, \Box(A \vee B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \vee \Diamond B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1, \Box A$	$\vee \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond B$	$\vee \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 4$
7.	$\mathbb{T} 1.1, A \vee B$	$\Box \mathbb{T} 2$
8.	$\mathbb{F} 1.1, B$	$\Diamond \mathbb{F} 5$
$\swarrow \quad \searrow$		
9.	$\mathbb{T} 1.1, A \quad \mathbb{T} 1.1, B$	$\vee \mathbb{T} 7$
	$\otimes \quad \otimes$	

This closes! Valid in K.

Example 9: $\Box A \rightarrow A$ in KT

We already did this one - it closes immediately with the $T\Box$ rule.

Example 10: $\Diamond \Box A \rightarrow A$ in KT

1. $\mathbb{F} 1, \Diamond \Box A \rightarrow A$ Assumption
2. $\mathbb{T} 1, \Diamond \Box A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, A \rightarrow \mathbb{F} 1$

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow A$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{T} 1.1, A$ | $\mathbb{T} \Box 4$ |

The tree stays open in KT - we can't get A back to world 1.

Example 11: $\Diamond \Box A \rightarrow A$ in KTB

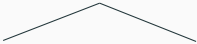
1.	$\mathbb{F} 1, \Diamond \Box A \rightarrow A$	Assumption
2.	$\mathbb{T} 1, \Diamond \Box A$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, \Box A$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\mathbb{T} \Box 4$
6.	$\mathbb{T} 1, A$	$\mathbb{B} \Box 5$
	$\otimes$	

With the $\mathbb{B}\Box$ rule, it closes!

Example 12: $\Box(A \wedge B) \leftrightarrow (\Box A \wedge \Box B)$ in K

Let's do just the left-to-right direction.

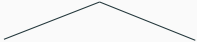
1. $\mathbb{F} 1, \Box(A \wedge B) \rightarrow (\Box A \wedge \Box B)$ Assumption
2. $\mathbb{T} 1, \Box(A \wedge B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box A \wedge \Box B \rightarrow \mathbb{F} 1$

1.	$\mathbb{F} 1, \Box(A \wedge B) \rightarrow (\Box A \wedge \Box B)$	Assumption
2.	$\mathbb{T} 1, \Box(A \wedge B)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A \wedge \Box B$	$\rightarrow \mathbb{F} 1$
		
4.	$\mathbb{F} 1, \Box A$ $\mathbb{F} 1, \Box B$	$\wedge \mathbb{F} 3$
5.	$\mathbb{F} 1.1, A$ $\mathbb{F} 1.1, B$	$\Box \mathbb{F} 4$
6.	$\mathbb{T} 1.1, A \wedge B$ $\mathbb{T} 1.1, A \wedge B$	$\Box \mathbb{T} 2$
7.	$\mathbb{T} 1.1, A$ $\mathbb{T} 1.1, B$	$\wedge \mathbb{T} 6$
	 	

Valid in K!

Example 13: $\Diamond(A \vee B) \leftrightarrow (\Diamond A \vee \Diamond B)$ in K

We already showed one direction. Let's do the other direction.

1.	$\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$	Assumption
2.	$\top 1, \Diamond A \vee \Diamond B$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(A \vee B)$	$\rightarrow \mathbb{F} 1$
		
4.	$\top 1, \Diamond A$	$\vee \top 2$
5.	$\top 1.1, A$	$\Diamond \top 4$
6.	$\mathbb{F} 1.1, A \vee B$	$\Diamond \mathbb{F} 3$
7.	$\mathbb{F} 1.1, A$	$\vee \mathbb{F} 6$
		

So the biconditional is valid in K!

Example 14: More Practice with T

$\Diamond A \rightarrow \Diamond\Diamond A$ in K vs KT

In K:

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond A \rightarrow \Diamond\Diamond A$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond\Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, A$ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{F} 1.1, \Diamond A$ | $\Diamond \mathbb{F} 3$ |

Stays open in K!

In KT:

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond A \rightarrow \Diamond\Diamond A$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond\Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, A$ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{F} 1.1, \Diamond A$ | $\Diamond \mathbb{F} 3$ |
| 6. | $\mathbb{F} 1.1, A$ | $\mathbb{T} \Diamond 5$ |



Closes in KT!

Example 15: Symmetry in Action

$$\Box A \rightarrow \Diamond A \text{ in K vs KB}$$

In K:

1. $\mathbb{F} 1, \Box A \rightarrow \Diamond A$ Assumption
2. $\mathbb{T} 1, \Box A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond A \rightarrow \mathbb{F} 1$

Stays open - world 1 might not access any worlds!

In KB:

1. $\mathbb{F} 1, \Box A \rightarrow \Diamond A$ Assumption
2. $\mathbb{T} 1, \Box A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond A \rightarrow \mathbb{F} 1$

Still stays open in KB alone! This needs reflexivity (T) to close.

In KT:

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Box A \rightarrow \Diamond A$ | Assumption |
| 2. | $\mathbb{T} 1, \Box A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, A$ | $\mathbb{T} \Box 2$ |
| 5. | $\mathbb{F} 1, A$ | $\mathbb{T} \Diamond 3$ |



Closes in KT!

Example 16: $\Diamond A \rightarrow \Box \Diamond A$ in S5

This is a characteristic theorem of S5 - let's prove it step by step using simplified S5.

1. $\models 1, \Diamond A \rightarrow \Box \Diamond A$ Assumption

Start with it being false at 1.

1. $\mathbb{F} 1, \Diamond A \rightarrow \Box \Diamond A$ Assumption
2. $\mathbb{T} 1, \Diamond A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box \Diamond A \rightarrow \mathbb{F} 1$

False conditional: true antecedent, false consequent.

- | | | |
|----|--|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond A \rightarrow \Box \Diamond A$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 2, A$ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{F} 3, \Diamond A$ | $\Box \mathbb{F} 3$ |

True $\Diamond$ sentences and false $\Box$ sentences have to be made true somewhere - not necessarily the same 'somewhere'.

1.	$\mathbb{F} 1, \Diamond A \rightarrow \Box \Diamond A$	Assumption
2.	$\mathbb{T} 1, \Diamond A$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box \Diamond A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 2, A$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{F} 3, \Diamond A$	$\Box \mathbb{F} 3$
6.	$\mathbb{F} 2, A$	$\Diamond \mathbb{F} 5$
		

False $\Diamond$ sentences are false everywhere - that is, the part inside the $\Diamond$ is false everywhere. So A is false at world 2. But we just said A is true at world 2! This closes, so it's a theorem of S5.

Example 17: $\Box\Diamond A \rightarrow A$ in S5

Let's test whether this is valid in S5. (Spoiler: it's not!)

1. $\models 1, \Box \Diamond A \rightarrow A$ Assumption

Start with it being false at 1.

1. $\mathbb{F} 1, \Box \Diamond A \rightarrow A$ Assumption
2. $\mathbb{T} 1, \Box \Diamond A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, A \rightarrow \mathbb{F} 1$

False conditional.

1. $\mathbb{F} 1, \Box \Diamond A \rightarrow A$ Assumption
2. $\mathbb{T} 1, \Box \Diamond A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, A \rightarrow \mathbb{F} 1$
4. $\mathbb{T} 1, \Diamond A \Box \mathbb{T} 2$

The unboxed part of a box sentence is true everywhere, so it is true here.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Box \Diamond A \rightarrow A$ | Assumption |
| 2. | $\mathbb{T} 1, \Box \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Diamond A$ | $\Box \mathbb{T} 2$ |
| 5. | $\mathbb{T} 2, A$ | $\Diamond \mathbb{T} 4$ |

True $\Diamond$ sentences have to be made true by some world. It can't be world 1, since A is false there.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Box \Diamond A \rightarrow A$ | Assumption |
| 2. | $\mathbb{T} 1, \Box \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Diamond A$ | $\Box \mathbb{T} 2$ |
| 5. | $\mathbb{T} 2, A$ | $\Diamond \mathbb{T} 4$ |
| 6. | $\mathbb{T} 2, \Diamond A$ | $\Box \mathbb{T} 2$ |

True $\Box$ sentences are true, unboxed, everywhere.

Now if we follow the rules mechanically we would add a new world 3 to make line 6 true, and the cycle would continue. But if we think through what the lines are saying, we can see that we can stop here:

- Line 6 ($\Diamond A$ true at 2) has to be made true somehow
- But line 5 says that world 2 itself makes line 6 true (since A is true at 2)
- So there isn't anything extra to do

So here is a model where this formula fails:

- There are two worlds: w_1 and w_2
- At w_1 , A is false
- At w_2 , A is true
- Because w_2 exists, $\Diamond A$ is true everywhere
- So $\Box \Diamond A$ is true everywhere
- But at w_1 , $\Box \Diamond A$ is true and A is false
- Therefore at w_1 , $\Box \Diamond A \rightarrow A$ is false

This shows that $\Box \Diamond A \rightarrow A$ is **not** valid in S5.

Example 18: $\Box(\Box A \rightarrow A) \rightarrow \Box A$ in KT4

1. $\mathbb{F} 1, \Box(\Box A \rightarrow A) \rightarrow \Box A$ Assumption
2. $\mathbb{T} 1, \Box(\Box A \rightarrow A) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box A \rightarrow \mathbb{F} 1$

1.	$\mathbb{F} 1, \Box(\Box A \rightarrow A) \rightarrow \Box A$	Assumption
2.	$\mathbb{T} 1, \Box(\Box A \rightarrow A)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A \rightarrow A$	$\mathbb{T} \Box 2$
5.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 3$
6.	$\mathbb{T} 1.1, \Box A \rightarrow A$	$\Box \mathbb{T} 2$

1.	$\mathbb{F} 1, \Box(\Box A \rightarrow A) \rightarrow \Box A$	Assumption
2.	$\mathbb{T} 1, \Box(\Box A \rightarrow A)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Box A \rightarrow A$	$\mathbb{T} \Box 2$
5.	$\mathbb{F} 1.1, A$	$\Box \mathbb{F} 3$
6.	$\mathbb{T} 1.1, \Box A \rightarrow A$	$\Box \mathbb{T} 2$
$\swarrow \quad \searrow$		
7.	$\mathbb{F} 1.1, \Box A \quad \mathbb{T} 1.1, A$	$\rightarrow \mathbb{T} 6$
8.	$\mathbb{F} 1.1.1, A \quad \otimes$	$\Box \mathbb{F} 7$
9.	$\mathbb{T} 1.1, A$	$\mathbb{T} \Box 6$
	$\otimes$	

Valid in KT4!

Example 19: $\Diamond\Box A \rightarrow \Box\Diamond A$ in S5

We'll use simplified S5 notation for this.

1. $\mathbb{F} 1, \Diamond \Box A \rightarrow \Box \Diamond A$ Assumption
2. $\mathbb{T} 1, \Diamond \Box A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Box \Diamond A \rightarrow \mathbb{F} 1$

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow \Box \Diamond A$ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 2, \Box A$ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{F} 3, \Diamond A$ | $\Box \mathbb{F} 3$ |

1.	$\mathbb{F} 1, \Diamond \Box A \rightarrow \Box \Diamond A$	Assumption
2.	$\mathbb{T} 1, \Diamond \Box A$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box \Diamond A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 2, \Box A$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{F} 3, \Diamond A$	$\Box \mathbb{F} 3$
6.	$\mathbb{T} 3, A$	$\Box \mathbb{T} 4$
7.	$\mathbb{F} 3, A$	$\Diamond \mathbb{F} 5$
	$\otimes$	

Closes in S5!

Example 20: $\neg \Box \neg A \rightarrow \Diamond A$ in K

1.	$\mathbb{F} 1, \neg \Box \neg A \rightarrow \Diamond A$	Assumption
2.	$\mathbb{T} 1, \neg \Box \neg A$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{F} 1, \Box \neg A$	$\neg \mathbb{T} 2$
5.	$\mathbb{F} 1.1, \neg A$	$\Box \mathbb{F} 4$
6.	$\mathbb{T} 1.1, A$	$\neg \mathbb{F} 5$
7.	$\mathbb{F} 1.1, A$	$\Diamond \mathbb{F} 3$
	$\otimes$	

Valid in K! This shows that $\neg \Box \neg A$ and $\Diamond A$ are equivalent.